Spatial and Temporal Analysis of Mangrove and Forest Change Based on Pre-Segmented Land Cover Data in Malang Regency

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***Abstract*—Effective monitoring of mangrove ecosystems is essential for their conservation and sustainable management. However, some regions may lack adequate remote sensing data due to factors such as temporal gaps, limited regional coverage, or suboptimal data quality, leading to reliance on available datasets. This study evaluates the Random Forest Classifier (RFC) for predicting land cover types across 16 coastal areas in Malang Regency using RGB-based land cover images. RFC achieved a high overall accuracy of up to 93% across three experiments, consistently demonstrating strong performance in identifying dominant classes, such as forest and mangroves, and outperforming the Convolutional Neural Network (CNN) model in accuracy. However, challenges were observed with minority classes, including forest loss and aquaculture ponds, reflecting the limitations of feature diversity and imbalanced data. These findings highlight the potential of RFC for analyzing land cover changes while emphasizing the importance of enhancing dataset quality and diversity to improve classification outcomes in underrepresented areas.**

Keywords—land cover, limited data, mangrove, random forest classifier.

I. Introduction

Mangroves are salt-tolerant forest ecosystems commonly found in tropical and subtropical coastal regions, along with halophytic vegetation such as *Suaeda*, *Salicornia*, and *Tamarix*. These plants have unique adaptations, including above-ground roots (*pneumatophores*), salt-excreting leaves, and floating seeds, enabling them to thrive in tidal environments with varying salinity, anaerobic soils, and high wind conditions [1]. Functioning as critical coastal ecosystems, mangroves reduce erosion, protect against extreme weather and high waves, and provide habitats for diverse species. They also enhance biodiversity, improve water quality, store substantial carbon, and play a vital role in supporting local communities by sustaining livelihoods and providing resources [2][3].

Despite their importance, mangrove ecosystems are facing significant threats. Global mangrove coverage has declined from 152,604 km² in 1996 to 147,359 km² in 2020, representing a 3.4% loss over 24 years [4]. This decline is primarily driven by human activities, such as land conversion, pollution from domestic, industrial, and oil discharges, as well as poorly managed coastal development [5]. Furthermore, climate change exacerbates these threats through rising sea levels, increased soil erosion, and extreme weather events, which further undermine the critical ecosystem services provided by mangroves.

Given the increasing threats to mangrove ecosystems, effective monitoring has become essential for their conservation and sustainable use [6]. Traditional field-based monitoring methods are resource-intensive, requiring significant time and cost [7]. As an alternative, remote sensing has emerged as a powerful tool for large-scale and long-term monitoring of mangrove ecosystems, offering the capability to accurately map distribution, assess vegetation density, and detect changes in physical conditions [8]. Furthermore, by conducting growth analysis and predicting mangrove expansion or reduction, remote sensing data can guide forest management strategies for sustainable mangrove preservation.

Remote sensing is widely used in mangrove monitoring, with satellite platforms like Sentinel-2 and Landsat 8 playing a significant role. These satellites provide rich spectral data across multiple bands, enabling the development and application of various spectral indices [9]. Indices such as NDVI, EVI, and NDWI are commonly used to enrich analysis by providing detailed information on ecosystem conditions and dynamics. However, in scenarios where access to complete satellite imagery is limited due to factors like temporal gaps, regional coverage issues, or poor data quality, the available data may be insufficient for comprehensive analysis. This limitation can significantly affect the accuracy and depth of the analysis, as it restricts the range of information needed for detailed evaluations, highlighting the challenges of working with incomplete or suboptimal data sources.

In this study, predictive analysis of mangrove forests and similar vegetation will be conducted using pre-segmented land cover images due to the limited availability of satellite imagery and the small amount of data in the study area. Unlike previous research that uses more detailed remote sensing data, the data in this case is restricted to RGB values, limiting the range of features that can be directly extracted. To overcome this limitation, feature engineering will be

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applied to enhance the data and maximize the number of useful features that can be derived.

II. RELATED WORKS

Recent studies have employed a variety of machine learning models to analyze trends, detect changes, and predict future outcomes in forest ecosystems, with Random Forest and deep learning models being widely used. Random Forest consistently demonstrated strong performance in various land cover classification and change detection tasks, often rivaling or even surpassing other machine learning models, particularly in scenarios with limited or relatively simple datasets.

Several studies have highlighted the effectiveness of Random Forest in different applications. A study on land cover change prediction in the Greater Amanzule Peatlands demonstrated the robustness of Random Forest in handling multi-source remote sensing data, achieving an overall accuracy of 85.3% and a Kappa coefficient of 0.81 [10]. Similarly, its application in urban land use and land cover changes using Landsat data resulted in an overall accuracy of 90.2% and a Kappa coefficient of 0.87, showcasing its capability in detecting urban expansion and land cover transformations [11]. In agricultural forecasting, specifically in predicting tea yields, Random Forest achieved a strong R value of 0.993, outperforming other models in predictive accuracy [12]. Moreover, in the context of mangrove forest analysis, Random Forest exhibited its highest performance, attaining an accuracy of 92.4% and a Kappa coefficient of 0.89 in classifying mangrove dynamics from Sentinel-2 time-series data [8]. These findings emphasize the versatility and adaptability of Random Forest across various environmental and agricultural applications.

However, when compared to deep learning models, Random Forest's performance can vary depending on the complexity of the task and the dataset size. For instance, CNN has been found to outperform Random Forest in forest classification tasks using high-resolution imagery, achieving an accuracy of 93.5% and an F1-score of 0.91. This higher accuracy is attributed to CNN ability to capture complex spatial patterns that Random Forest may miss in more intricate classification problems [13]. Similarly, deep learning models used for mangrove change detection in Myanmar achieved an accuracy of 94.1% and an Intersection over Union (IoU) score of 0.83, surpassing Random Forest in terms of feature detection and classification accuracy in complex scenarios[6].

While deep learning models excel in capturing intricate patterns and high-resolution data, Random Forest remains a reliable option for scenarios with limited data and computational resources. Studies indicate that Random Forest is frequently applied to smaller datasets with fewer features, whereas deep learning models require larger and more complex datasets for optimal performance. In this study, Random Forest is chosen due to the relatively small dataset and the simplicity of available RGB features compared to multi-band data typically used in deep learning. Notably, research has shown that Random Forest can achieve high accuracy even with a limited number of features, as demonstrated in a study on tea yield prediction [12], highlighting its effectiveness in feature-constrained environments.

III. METHODOLOGY

An overview of the methodology employed in this study is shown in Fig. 1.

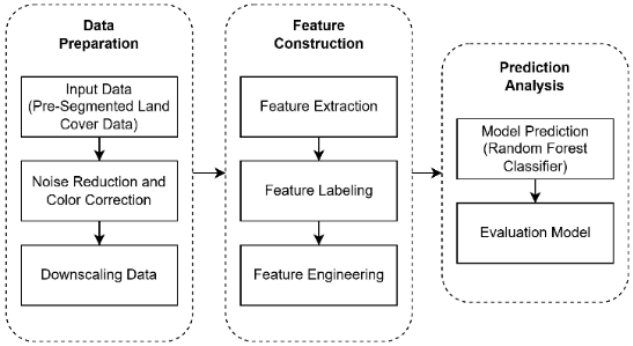

Fig. 1. Overview of the methodology used in this study.

A. Data Definition

In this study, the dataset was obtained from the digitization of Google Earth imagery spanning the years 2018 to 2023 and administrative boundary data from Global Administrative Areas (GADM) Indonesia 2019. The study area is geographically bounded by the coordinates between longitude 112.0371° and 112.9468° and latitude -7.2299° and -8.4021°, covering 16 selected coastal areas in Malang Regency. These areas include Bajul Mati Beach, Balekambang Beach, Bantol Beach, Goa Cina Beach, Jembatan Panjang Beach, Jongring Saloko Beach, Kondang Iwak Beach, Kondang Merak Beach, Licin Beach, Modangan Beach, Ngliyep Beach, Sempu Beach, Sendang Biru Beach, Sipelot Beach, Tamban Beach, and Wonogoro Beach.

The dataset consists of pre-segmented land cover data with a primary focus on vegetation dynamics in the selected areas. The imagery data was processed into square tiles with a resolution of 590x590 pixels, represented using the RGB color model to classify different land cover types. Each color corresponds to a specific land cover class: green represents forested areas, red indicates forest loss, blue denotes water bodies such as the sea, yellow represents residential areas, and cyan indicates aquaculture ponds.

B. Data Preparation

To ensure data quality and reliability, a comprehensive preparation process was conducted to address cloud cover and noise issues in raw land cover images. Cloud cover can obscure imagery, leading to inaccuracies, which were mitigated using noise reduction techniques. A spatial filtering method with mean filtering was applied to identify and smooth noisy pixels, particularly those affected by cloud cover. This approach enhances image clarity and ensures accurate land cover representation, preserving data integrity for further analysis. Studies have demonstrated the effectiveness of such filtering methods in enhancing image clarity and supporting more accurate representation of the underlying land cover, helping to preserve the integrity of the data for further analysis [14].

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Following the noise reduction, color correction was applied to standardize RGB values across the dataset, ensuring consistent pixel representation. Aligning RGB values to a uniform scale improved land cover segmentation by enhancing contrast and reducing misclassification risks. These preprocessing steps were crucial for ensuring the reliability and accuracy of the dataset during the predictive modeling phase. A comparison of the raw data and the corrected data is presented in Fig 2.

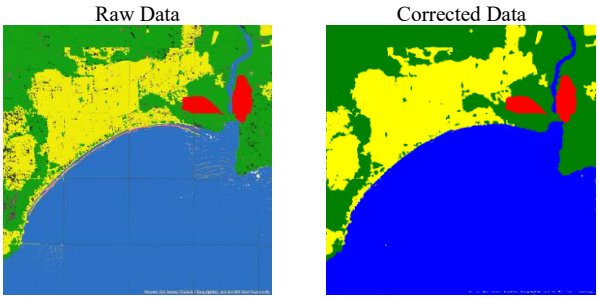

Fig. 2. Comparison of raw and pre-processed land cover images on Pantai Tamban 2018.

To optimize the dataset, the original 590x590 pixel land cover images were downscaled to 59x59 pixels using the nearest-neighbor interpolation method. This technique was chosen for its simplicity and ability to maintain the original pixel values without introducing noise or distortions. The downscaling process significantly reduced dataset size, enhancing computational efficiency while preserving essential spatial information for analysis as shown in Fig. 3.

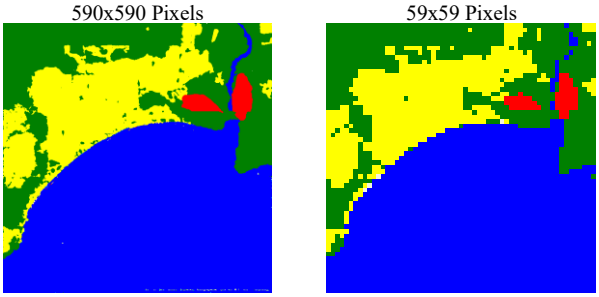

Fig. 3. Comparison of downscaled image on Pantai Tamban 2018.

C. *Feature Construction*

Feature selection in this study focused on extracting and engineering relevant features from the pre-processed land cover images to maximize the information available for analysis. The primary features extracted were the RGB values of each pixel, which represent the intensity of red, green, and blue colors in the image. These RGB values were then categorized into specific labels based on predefined thresholds, as presented in Table I.

TABLE I. LABELED RGB VALUE

Red Value	Green Value	Blue Value	Description	Color Name	Label
255	0	0	Forest Loss	Red	0
0	128	0	Forest	Green	1
0	0	255	Water Bodies	Blue	2
255	255	0	Residential Areas	Yellow	3
0	255	255	Aquaculture Ponds	Cyan	4

However, during the data processing stage, certain pixels with colors outside the predefined classifications were observed in images that did not represent any valid land cover class. These pixels were residues from the noise reduction process that had not entirely removed visual disturbances in the image data. As a result, an additional label, Class 5, was introduced to represent pixels that did not align with the defined land cover categories

To enrich the dataset further, an additional feature was engineered by incorporating spatial information from neighboring pixels. This technique employed a 3x3 window, where each pixel was analyzed alongside its 8 neighboring pixels to incorporate contextual information around each pixel, enhancing the model's ability to detect patterns and relationships between land cover classes. Another feature was also introduced by detecting pixel changes over time, identifying whether a pixel had undergone a change in class between the previous year and the current year.

Following the extraction and spatial feature engineering, the data were transformed into a structured dataset. Each record within the dataset represents a single pixel from a specific coastal area, with its position defined by the x-axis and y-axis coordinates. To ensure comprehensive representation, a feature labeled Area was included to specify the coastal region where the pixel was located. The dataset also includes temporal features representing the years 2018, 2019, 2020, 2021, 2022, and 2023, where the value in each year indicates the land cover category of that pixel based on the predefined classification labels.

To further expand the dataset, the neighboring pixel feature engineering was applied across all temporal features. Each year's feature set was enriched by including the pixel values of its 8 neighboring pixels as additional features. Additionally, a pixel change detection feature was added, providing temporal insights into dynamic land cover transformations. Consequently, the final structured dataset integrated spatial, temporal, and categorical information to provide a comprehensive representation of the land cover patterns across all coastal areas under study.

TABLE II. SAMPLE OF STRUCTURED DATASET

x	y	Area	2018	2019		…	NP1_2018	NP1_2019	…	Diff_2018_2019	Diff_2019_2020	…
0	0	Tamban	1	1		…	1	3	…	1	1	…
0	1	Tamban	1	1		…	4	4	…	4	4	…
0	2	Tamban	1	0		…	1	1	…	1	1	…

Table II contains the following features where x and y represent the coordinates of each pixel, Area denotes the coastal location of the pixel, year columns like 2018 and 2019 indicate the land cover class for that pixel during the respective year, NP columns including NP1_2018 and NP1_2019 represent the values of the 8 surrounding neighboring pixels for each year to include spatial context,

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and Diff columns including Diff_2018_2019 and Diff_2019_2020 indicate yearly changes in the land cover class, with a value of 1 if a change occurred between consecutive years and 0 if there was no change.

D. Model Construction

The model used in this study is the Random Forest Classifier (RFC), a robust ensemble learning method that combines multiple decision trees to improve classification performance and reduce data correlation [11]. This is achieved by incorporating two levels of randomness during model construction: random selection of subsets of data through bootstrap sampling and random selection of subsets of features at each decision tree split. This dual randomization not only reduces the correlation between individual trees but also minimizes the risk of overfitting, leading to a more accurate and reliable model.

The general process for constructing a Random Forest involves assembling and trees decision trees, where each tree is trained on a bootstrap sample of the data. The prediction from each tree is then aggregated to provide the final output. The formula for the prediction from a Random Forest is as shown in (1)

$$y = \frac{1}{B}\sum_{b=1}^{B} t_b(x) \tag{1}$$

Where B is the number of trees and $t_b(x)$ is s the prediction from the b-th tree for the input x.

One of Random Forest's key strengths is its ability to manage datasets with limited observations and features, which is particularly important for this study's mangrove land cover analysis. Unlike many complex models that require large and rich datasets to perform effectively, Random Forest is highly adaptable to small datasets. Its ensemble approach and aggregation mechanism ensure robust predictions, even in resource-constrained settings. Research has demonstrated that Random Forest can achieve strong classification accuracy and maintain predictive performance even when data availability is restricted [15].

E. Evaluation

In this study, the performance of Random Forest model will be evaluated using a range of standard classification metrics to assess its effectiveness in predicting land cover data. The evaluation will focus on Overall Accuracy (OA), Precision, Recall, and F1-Score, all of which are widely used in classification tasks to ensure a robust assessment of model performance. Overall Accuracy (OA) measures the proportion of correctly classified pixels out of the total samples. The formula for OA is shown in (2)

$$OA = \frac{TP + TN}{TP + TN + FP + FN} \tag{2}$$

Where TP is True Positives, TN is True Negatives, FP is False Positives, and FN is False Negatives.

Precision evaluates the model's ability to minimize false positives, indicating the accuracy of positive predictions as shown in (3)

$$Precision = \frac{TP}{TP + FP} \tag{3}$$

Recall measures the model's ability to capture true positives, indicating its sensitivity. The formula for Recall is shown in (4)

$$Recall = \frac{TP}{TP + FN} \tag{4}$$

F1-Score combines Precision and Recall into a single metric, providing a balanced evaluation, especially for imbalanced datasets as mentioned in (5)

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{5}$$

IV. Result and Discussion

A. Experiment Set Up

This study utilizes Random Forest Classifier (RFC) and Convolutional Neural Network (CNN) for land cover prediction, with CNN serving as a comparative model to assess RFC's effectiveness. Both models were tested on pre-segmented land cover images from 16 coastal areas in Malang Regency. After preprocessing, data from 13 areas were used for training, while data from Sipelot Beach, Tamban Beach, and Wonogoro Beach were reserved for testing. Model performance was evaluated through three experiments: predicting land cover for 2021 using data from 2018 to 2020, predicting 2022 using data from 2018 to 2021, and predicting 2023 using data from 2018 to 2022.

The Random Forest Classifier (RFC) was trained using 200 decision trees to ensure sufficient diversity among the individual classifiers. To mitigate overfitting, the maximum depth was limited to 5, preventing overly complex decision boundaries that could compromise generalization. Additionally, the model used the sqrt strategy for feature selection, which randomly selects a subset of features at each split to reduce correlation between trees, further improving the model's robustness and stability. The random state was set to 42 to ensure reproducibility.

For comparison, a Convolutional Neural Network (CNN) was developed using a Conv1D layer with 64 filters and a kernel size of 3, followed by MaxPooling1D to reduce dimensionality. Another Conv1D layer with 128 filters was added for deeper feature extraction. The output was flattened and passed through a Dense layer with 128 units and ReLU activation before the final softmax output layer. The model was compiled using the Adam optimizer and sparse categorical cross-entropy as the loss function.

B. Result

Following the preprocessing and feature engineering stages, both the Random Forest Classifier (RFC) and Convolutional Neural Network (CNN) models were trained and tested using the prepared dataset as described in the experiment setup. The results for the experiments are summarized in Table III.

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TABLE III. EXPERIMENT EVALUATION

Model	OA	Precision	Recall	F-1 Score
RFC 2021	**93.7%**	**72%**	**80%**	**76%**
RFC 2022	93.6%	70%	73%	71%
RFC 2023	93.4%	66%	78%	67%
CNN 2021	93,4%	62%	80%	65%
CNN 2022	93,4%	66%	74%	65%
CNN 2023	93,3%	66%	79%	67%

According to the evaluation results, the Random Forest Classifier (RFC) and Convolutional Neural Network (CNN) models both demonstrated consistent performance in all three experiments, indicating their capacity to manage tasks involving the classification of land cover. To guarantee equitable consideration for all classes, including minority categories, the evaluation metrics were computed using macro-averaged values. These findings offer a thorough comprehension of each model's generalization across various temporal datasets.

Overall, RFC consistently outperformed CNN in most metrics across all experiments, particularly in precision and F1-score. In the 2021 experiment, where the dataset was the smallest due to a shorter training range, RFC achieved the highest overall accuracy of 93.7% and the best F1-score of 76%, showcasing its ability to generalize effectively even with reduced data. In the 2022 and 2023 experiments, RFC maintained high performance, with accuracy slightly decreasing to 93.6% and 93.4%, respectively, but still surpassing CNN. While CNN achieved similar overall accuracy scores, its consistently lower precision resulted in weaker F1-scores, indicating its higher tendency to produce false positives.

To further illustrate the Random Forest Classifier model's performance, Fig. 4, Fig. 5, and Fig. 6 provide a visual comparison of the predicted and actual land cover images for Sipelot Beach, Tamban Beach, and Wonogoro Beach, respectively, based on predictions made for the year 2023.

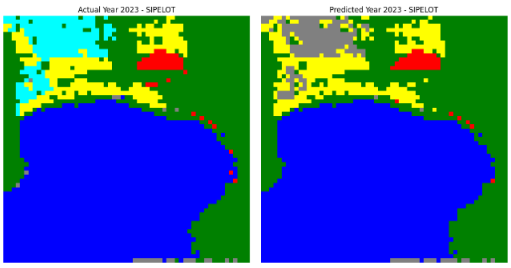

Fig. 4. Actual and predicted image of Sipelot Beach.

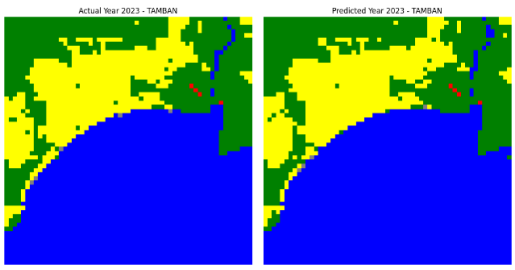

Fig. 5. Actual and predicted image of Tamban Beach.

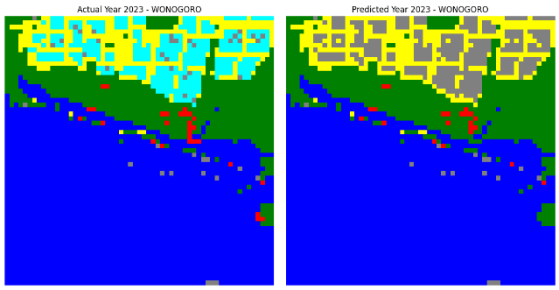

Fig. 6. Actual and predicted image of Wonogoro Beach.

The model performed well overall, achieving a high average accuracy with relatively few false predicted pixels, as indicated by the numerical results. However, when examining the visual comparisons, it becomes clear that there are areas where the model's predictions are less accurate. Specifically, it is clear from the images that the model struggled to accurately classify land cover areas with Class 4, representing aquaculture ponds. In contrast, for the other classes, the visualizations show that the predicted and actual land cover images closely align, with only minor differences visible.

To gain deeper insight into the model's classification performance, a confusion matrix was generated. The confusion matrix shown below in Fig. 7 provides detailed information about the Random Forest Classifier's predictions.

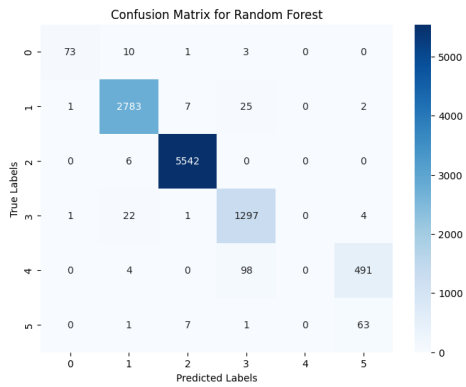

Fig. 7. Confusion matrix.

From the confusion matrix, metrics such as accuracy, precision, recall, and F1-score provide valuable insights into the model's performance in classifying each category. For Class 1, representing the forest, the model demonstrates excellent performance with a precision of 98%, a recall of 99%, and an F1-score of 99%. According to the confusion matrix, the model successfully identified 2,783 true positives, accurately classifying forest instances. The number of false positives, which are instances incorrectly predicted as forest, amounts to 43 cases. Additionally, false negatives, representing forest instances misclassified as other categories, amount to 35 cases. These results highlight the model's exceptional ability to recognize and accurately predict the forest category with minimal error

Despite the model's strong overall performance, certain limitations are evident, particularly in Class 0, which represents forest loss, and Class 4, which represents aquaculture ponds. For Class 0, the model achieves a high

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precision of 97%, indicating that most predictions labeled as forest loss are correct. Out of 87 samples, 73 were correctly identified as true positives, while 14 instances were misclassified as other categories. The recall indicates that nearly 16% of forest loss instances remain undetected, which could lead to significant misinterpretation. The performance for Class 4, representing aquaculture ponds, is significantly weaker. The model achieves a precision and recall of 0%, meaning none of the 593 actual instances were correctly classified, and all were misclassified into other categories. This severe misclassification indicates that the features for aquaculture ponds lack distinctiveness or overlap heavily with other classes, making it challenging for the model to differentiate this class.

C. Discussion

The results from the Random Forest Classifier indicate that while the model demonstrated high performance with an overall accuracy, it faced significant challenges with specific land cover classes, particularly Class 0 and Class 4. Possible reasons for these misclassifications include the limited variation in features that could be extracted, as the dataset relied solely on RGB-based features. This limitation likely constrained the model's ability to effectively differentiate between certain categories. Additionally, imbalanced data distribution may have contributed to the poor performance of these classes, as fewer samples can lead to reduced sensitivity and hinder the model's ability to generalize effectively. Addressing this issue requires resampling techniques to balance the class distribution. Equal representation of all classes would enable the model to better capture the unique characteristics of underrepresented. Combining these adjustments with advanced feature engineering or incorporating multispectral data could further enhance the model's capacity to differentiate between challenging classes.

V. CONCLUSION

This study aimed to predict mangrove land cover and its changes using a pre-segmented land cover dataset with limited features based on RGB values. To address these limitations, feature engineering was applied to enhance data quality and improve model performance. The Random Forest Classifier achieved an overall accuracy of up to 93% across three experiments, demonstrating strong performance in identifying forest areas with high precision and recall. However, challenges arose in detecting forest loss and aquaculture ponds, with lower recall for forest loss indicating difficulties in identifying fragmented areas, and severe misclassification of aquaculture ponds highlighting the limitations of RGB-based features possibility of imbalanced data distribution. These issues reduced the model's sensitivity and generalization for certain classes.

To overcome the observed challenges, future research could address these limitations by incorporating diverse data sources, such as multispectral imagery, to enhance feature extraction and class differentiation. Resampling techniques can balance class distribution and improve the detection of underperforming classes. Exploring advanced machine learning models or hybrid approaches may further refine accuracy and handle complex land cover patterns, strengthening the use of Random Forest for reliable mangrove ecosystem monitoring.

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